MEESHO WOMEN'S FASHION DATABASE MANAGEMENT SYSTEM

DATABASE RELATIONSHIP ANALYSIS USING JOINS

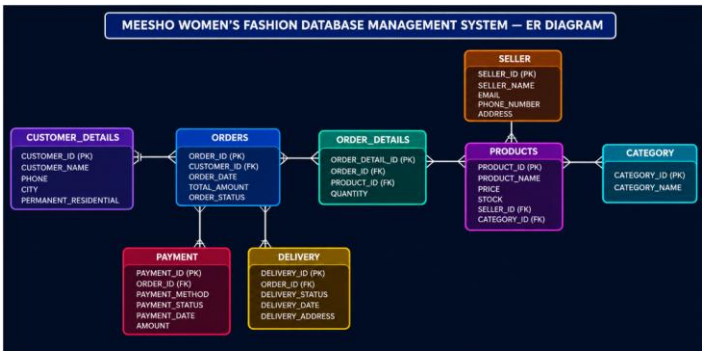
DBMS – WEEK 8

Objectives

1. Combine Customer, Product, Orders, Order_Details, and Payments tables.
2. Implement INNER JOIN, LEFT JOIN, and RIGHT JOIN.
3. Retrieve complete order details.
4. Display customer purchase history.
5. Generate multi-table and payment transaction reports.

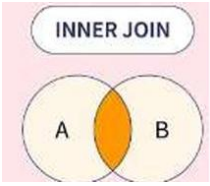
1. Combining Multiple Tables

The Meesho Women's Fashion database contains different tables that store customer, product, order, and payment information. These tables can be combined using JOIN operations to retrieve related information.



2. INNER JOIN

INNER JOIN returns only the records that have matching values in both tables.



2.1 Customer and Orders

```
SQL> SELECT C.CUSTOMER_ID, C.CUSTOMER_NAME,
  2         O.ORDER_ID, O.ORDER_DATE, O.ORDER_STATUS
  3  FROM CUSTOMER C
  4  INNER JOIN ORDERS O
  5  ON C.CUSTOMER_ID = O.CUSTOMER_ID;

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------
  ORDER_ID ORDER_DAT ORDER_STATUS
---------- --------- ------------------------------
       101
Priya
       201 10-SEP-26 Delivered

       102
Divya
       202 11-SEP-26 Shipped
```

This displays customers who have placed orders.

2.2 Orders and Payment



```
SQL> SELECT O.ORDER_ID,
  2         P.PAYMENT_ID,
  3         P.PAYMENT_METHOD,
  4         P.AMOUNT,
  5         P.PAYMENT_STATUS,
  6         P.PAYMENT_DATE
  7  FROM ORDERS O
  8  INNER JOIN PAYMENT P
  9  ON O.ORDER_ID = P.ORDER_ID;

  ORDER_ID PAYMENT_ID PAYMENT_METHOD                    AMOUNT
---------- ---------- ------------------------------ ----------
PAYMENT_STATUS                 PAYMENT_D
------------------------------ ---------
       201        401 UPI                                   899
Success                        10-SEP-26

       202        402 Credit Card                          1299
Success                        11-SEP-26

       203        403 Debit Card                            599
Success                        12-SEP-26


  ORDER_ID PAYMENT_ID PAYMENT_METHOD                    AMOUNT
---------- ---------- ------------------------------ ----------
PAYMENT_STATUS                 PAYMENT_D
------------------------------ ---------
       204        404 UPI                                  1499
```

This displays orders and their corresponding payment details.

2.3 Product and Order_Details

```
SQL> SELECT P.PRODUCT_ID,
  2         P.PRODUCT_NAME,
  3         OD.ORDER_ID,
  4         OD.QUANTITY
  5  FROM PRODUCTS P
  6  INNER JOIN ORDER_DETAILS OD
  7  ON P.PRODUCT_ID = OD.PRODUCT_ID;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------
  ORDER_ID   QUANTITY
---------- ----------
       101
Laptop
       201          2

       101
Laptop
       205          1

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------
  ORDER_ID   QUANTITY
---------- ----------
```

This displays products that have been included in orders.

3. LEFT JOIN

LEFT JOIN returns all records from the left table and matching records from the right table.

3.1 Display All Customers and Their Orders



```
SQL> SELECT C.CUSTOMER_ID,
  2         C.CUSTOMER_NAME,
  3         O.ORDER_ID
  4  FROM CUSTOMER_DETAILS C
  5  LEFT JOIN ORDERS O
  6  ON C.CUSTOMER_ID = O.CUSTOMER_ID;

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------
  ORDER_ID
----------
       101
Anitha
       201

       101
Anitha
       211

CUSTOMER_ID
-----------
CUSTOMER_NAME
```

This displays all customers, including customers who have not placed any orders.

4. RIGHT JOIN

RIGHT JOIN returns all records from the right table and matching records from the left table.

4.1 Display All Orders and Customer Details

```
SQL> SELECT C.CUSTOMER_ID,
  2         C.CUSTOMER_NAME,
  3         O.ORDER_ID
  4  FROM CUSTOMER_DETAILS C
  5  RIGHT JOIN ORDERS O
  6  ON C.CUSTOMER_ID = O.CUSTOMER_ID;

CUSTOMER_ID
-----------
CUSTOMER_NAME
----------------------------------------
  ORDER_ID
----------
         101
Anitha
       201

         102
Priya
       202

CUSTOMER_ID
-----------
CUSTOMER_NAME
----------------------------------------
  ORDER_ID
----------
```

This displays all orders along with matching customer information.

5. Complete Order Details

This query combines Customer, Orders, Order_Details, Product, and Payment tables to retrieve complete order information.

UCT_ID

LEFT JOIN PAYMENTS PAY

ON O.ORDER_ID = PAY.ORDER_ID;

6. **Customer Purchase History**

```
SQL> SELECT C.CUSTOMER_NAME,
  2         O.ORDER_ID,
  3         P.PRODUCT_NAME,
  4         OD.QUANTITY
  5  FROM CUSTOMER_DETAILS C
  6  JOIN ORDERS O
  7  ON C.CUSTOMER_ID = O.CUSTOMER_ID
  8  JOIN ORDER_DETAILS OD
  9  ON O.ORDER_ID = OD.ORDER_ID
 10  JOIN PRODUCTS P
 11  ON OD.PRODUCT_ID = P.PRODUCT_ID;

CUSTOMER_NAME
----------------------------------------
  ORDER_ID
----------
PRODUCT_NAME
----------------------------------------
  QUANTITY
----------
Anitha
       201
Laptop
         2
```

This query displays the products purchased by each customer.

7. Customer Purchase History with Payment

This query displays customer purchase details along with payment amount and status.

```
SQL> SELECT C.CUSTOMER_ID,
  2         C.CUSTOMER_NAME,
  3         O.ORDER_ID,
  4         P.PRODUCT_NAME,
  5         OD.QUANTITY
  6  FROM CUSTOMER_DETAILS C
  7  JOIN ORDERS O
  8  ON C.CUSTOMER_ID = O.CUSTOMER_ID
  9  JOIN ORDER_DETAILS OD
 10  ON O.ORDER_ID = OD.ORDER_ID
 11  JOIN PRODUCTS P
 12  ON OD.PRODUCT_ID = P.PRODUCT_ID
 13  ORDER BY C.CUSTOMER_ID;

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------
  ORDER_ID
----------
PRODUCT_NAME
--------------------------------------------
  QUANTITY
----------
        101
Anitha
       201

CUSTOMER_ID
-----------
```

8. Multi-Table Business Report

This query generates a customer purchase report showing the total number of orders and total purchase amount for each customer.

```
SQL> SELECT C.CUSTOMER_ID,
  2         C.CUSTOMER_NAME,
  3         COUNT(DISTINCT O.ORDER_ID) AS TOTAL_
  4  FROM CUSTOMER_DETAILS C
  5  LEFT JOIN ORDERS O
  6  ON C.CUSTOMER_ID = O.CUSTOMER_ID
  7  LEFT JOIN ORDER_DETAILS OD
  8  ON O.ORDER_ID = OD.ORDER_ID
  9  LEFT JOIN PRODUCTS P
 10  ON OD.PRODUCT_ID = P.PRODUCT_ID
 11  GROUP BY C.CUSTOMER_ID, C.CUSTOMER_NAME
 12  ORDER BY TOTAL_ORDERS DESC;

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------
TOTAL_ORDERS
------------
        101
Anitha
           2

        102
Priya
           1

CUSTOMER_ID
```

9. Payment Transaction Report

This query displays payment transaction details along with the related order and customer information.

```
SQL> SELECT PAY.PAYMENT_ID,
  2         O.ORDER_ID,
  3         C.CUSTOMER_NAME,
  4         PAY.PAYMENT_METHOD,
  5         PAY.AMOUNT,
  6         PAY.PAYMENT_STATUS
  7  FROM PAYMENT PAY
  8  JOIN ORDERS O
  9  ON PAY.ORDER_ID = O.ORDER_ID
 10  LEFT JOIN CUSTOMER_DETAILS C
 11  ON O.CUSTOMER_ID = C.CUSTOMER_ID
 12  ORDER BY PAY.PAYMENT_ID;

PAYMENT_ID   ORDER_ID
---------- ----------
CUSTOMER_NAME
--------------------------------------------------------------------
PAYMENT_METHOD                   AMOUNT PAYMENT_STATUS
-------------------------- ---------- --------------------------
       401        201
Anitha
UPI                                899 Success

       402        202
Priya
Credit Card                       1299 Success

PAYMENT_ID   ORDER_ID
```

```
SQL> DESC CUSTOMER;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 CUSTOMER_ID                               NOT NULL NUMBER(38)
 CUSTOMER_NAME                             NOT NULL VARCHAR2(100)
 EMAIL                                     NOT NULL VARCHAR2(100)
 PHONE_NUMBER                              NOT NULL VARCHAR2(15)
 ADDRESS                                   NOT NULL VARCHAR2(255)
```

Conclusion

JOIN operations help combine related information from multiple tables in the Meesho Women's Fashion Database Management System. They are useful for retrieving complete order details, tracking customer purchase history, viewing payment information, and generating business reports.